

Clustering Districts by Educational Performance Using Machine Learning and Neutrosophic Logic

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Abstract. Educational performance evaluation at the district level consists of several indicators that are dependent on each other and in many cases they are structurally imbalanced. Traditional clustering methods typically operate in normalized Euclidean feature spaces. They potentially overlook intra-district instability and peer-relative performance gaps. The proposed study suggests a statistically based neutrosophic feature transformation to cluster the dataset of 768 Indian districts with 6 educational dimensions, the Performance Grading Index-District (PGI-D) 2022-23 & 2023-24. The independent Truth (normalized achievement), Falsity (peer-relative deficit) and Indeterminacy (intra-district structural imbalance) components are used to represent each district. This transformation extends the classical feature space into an uncertainty-aware representation. Comparative clustering is done on the basis of Euclidean K-means and the cosine based agglomerative algorithms. Although the Euclidean separability is not much different, $\approx 46\%$ of the districts are reassigned under the neutrosophic transformation. In contrast, cosine distance based clustering demonstrates significant growth in silhouette score (0.269 to 0.389), which means that it provides a better pattern-level discrimination. The findings indicate that neutrosophic feature expansion performance is geometry-specific, and it is crucial to match uncertainty modeling and clustering geometry in multi-criteria educational analytics.

Keywords: Neutrosophic Logic · Educational Analytics · Multi-Criteria Clustering · Cosine Distance · Uncertainty Modeling · Geometry-Specific Clustering · PGI-D

1 Introduction

In big and diverse countries such as India, statistical assessment of education systems has become mainstream for the government especially when it comes to planning of policies and resource distribution. District-level assessment systems, including the Performance Grading Index-District (PGI-D), combine a vast variety of indicators like learning results, the quality of teaching, infrastructure and digital preparedness [1]. These multi-criteria datasets are heterogeneous by nature and often have structural imbalances, i.e. they are unequally

developed socio-economically and have uneven institutional capacity in different regions [2]. Although the PGI-D 2022-23 & 2023-24 framework offers a rich, multi-dimensional standardized assessment of 768 Indian districts, clustering the districts into meaningful performance clusters is a major methodological challenge.

The old-fashioned clustering algorithms like K-means use normalized Euclidean feature spaces, and they presuppose magnitude-based similarity and cluster shape, which is nearly spherical [3, 4]. Performance indicators in high-dimensional learning data tend to be correlated and heterogeneously dispersed. Euclidean distance in such datasets cannot adequately represent structural imbalance between dimensions. a district can have excellent infrastructure but poor learning results or digital connectivity, which causes intra-district instability that cannot be reflected by magnitude-based similarity. Additionally, distance measures become concentrated as the dimensionality rises and hence lose their power to discriminate, a phenomenon called curse of dimensionality [5]. These shortcomings drive the development of alternative similarity metrics and uncertainty-sensitive representations to more general Euclidean clustering.

In the multi-criteria decision analysis, uncertainty modeling has gained more prominence. Fuzzy set theory proposes graded membership and usually restricts a truth value and a false value by complement [6].

$$X(T, F) : T(Truth) = 1 - F(Falsity) \therefore \text{Fuzzy membership function.}$$

Neutrosophic set theory goes further to model Truth (T), Indeterminacy (I), and Falsity (F) independently and does not construct the algebraic dependency between these three membership functions. [7].

$$N(T, I, F) : T(Truth) \neq 1 - F(Falsity) \therefore \text{Neutrosophic membership function.}$$

Among educational analytics, Truth, Falsity and Indeterminacy when applied to educational analytics could be the normalized achievement of a district, its peer-relative deficit, and the structural imbalance within the various facets of education, respectively. This triple offers a more detailed model which provides the picture of the fluidity of the real-world educational performance where fuzzy or binary sets may oversimplify it.

Contributions to this work are thus:

1. A data-driven construction of neutrosophic features for educational analytics at the district level.
2. A numerical analysis of neutrosophic expansion behavior under varying geometric configurations.
3. Empirical evidence demonstrating that pattern-level separability improves significantly when uncertainty modeling is aligned with cosine similarity, thereby providing a more sensitive framework for analyzing structural responses to educational policy.

2 Literature Survey

This literature survey was executed to understand the current geometric-specific clustering and to extend its applications towards the PGI-D for achieving better and uncertainty aware clustering of districts.

Table 1. Comprehensive Summary of Related Work

Author (Year)	Category	Method	Geometry	Core Focus
Jain (2010) [4]	Survey	Clustering Review	Mixed	Taxonomy of clustering paradigms.
Xu & Tian (2015) [8]	Survey	Algorithm Survey	Mixed	Categorization by geometry and structure.
Ye (2014) [9]	Neutrosophic	SVNS Clustering	Similarity-based	Neutrosophic similarity-based clustering.
Qiu et al. (2024) [10]	Neutrosophic	N-FCM	Euclidean+ Uncertainty	Boundary indeterminacy modeling.
Smarandache (2025) [11]	Neutrosophic Theory	Neutrosophication	Data-driven	Comparative T,I,F construction methods.
Mandour et al. (2025) [12]	Neutrosophic	T,I,F Modeling	High-dimensional	Conflict handling in multi-criteria data.
Afchal et al. (2025) [13]	Neutrosophic	Academic Prediction	Uncertainty-aware	Performance modeling under ambiguity.
Romero & Ventura (2010) [14]	Education	EDM Survey	Mixed	AI techniques in educational data mining.
Baker & Inven-tado (2016) [15]	Education	Learning Analytics	Mixed	Data-driven educational analytics framework.
Obadi (2010) [16]	Education	Spectral (Education)	Graph-based	Pattern grouping beyond magnitude similarity.
Sultan (2023) [17]	Education	K-means (Education)	Euclidean	Student/institution performance clustering.
Namoun & Al-shanqiti (2020) [2]	Education	Heterogeneous Data	Non-uniform	Clustering challenges in diverse national systems.
Venkatesan & Mappillairaju (2023) [18]	Spatial	Spatial Clustering	Spatial geometry	Spatial correlation in Indian education data.
Ministry of Education (2024) [1]	Dataset	PGI-D Framework	Multi-criteria	District evaluation structure.

3 Methodology

The suggested framework is a hybrid approach of feature modeling based on data and the geometry-informed clustering of educational performance at the district level. Figure 1 shows the methodological workflow. This analysis applies the Performance Grading Index-District (PGI-D) dataset 2022-23 and 2023-24, which was published by the Ministry of Education, Government of India [1]. The dataset scores 768 districts in six educational variables, namely, Outcome, Effective Classroom Teaching (ECT), Infrastructure and Facilities (IF&SE), School Safety and Child Protection (SS&CP), Digital Learning (DL), and Governance Practices (GP).

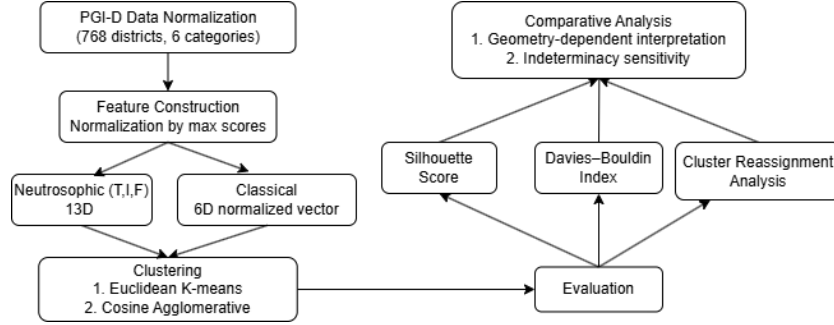


Fig. 1. Geometry-aware neutrosophic clustering framework for PGI-D district analysis.

1. Exploratory Data Analysis

Exploratory examination of the PGI-D data showed unequally dispersed categories. Specifically, Digital Learning had significantly higher coefficient of variation than Outcome, and it suggests structural imbalance among indicators. In addition, a correlation analysis revealed moderate levels of interdependence between the governance and infrastructure-related dimensions. These results indicate that the similarity of the districts cannot be fully described by magnitude, which encourages the representation of uncertainty. The correlation heatmap is given in Figure 2. These findings suggest that educational performance in districts is not only based on magnitude but is also characterized by structural imbalance of indicators. Districts can be strong in terms of infrastructure and weak in terms of digital

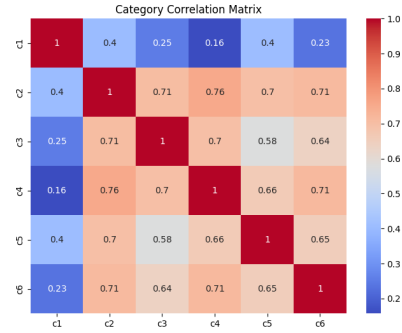


Fig. 2. Category Correlation Matrix

learning or governance. Traditional normalized feature vectors cannot be fully used to capture such internal inconsistency. In turn, both structural instability and performance strength need to be modeled with the help of uncertainty-aware representation.

2. Neutrosophic Feature Construction:

In order to represent these attributes, the data is converted to neutrosophic representation, which comprises Truth (T), Falsity (F), and Indeterminacy (I) elements. These three membership functions in neutrosophic set theory are independent and are not limited by the correlation: $T + I + F = 1$ [7].

All the components are aimed to reflect a particular statistical aspect of district performance.

Truth (T): Strength of Performance.

Truth is the standardized level of attainment in a district of a specific dimension of education. The maximum score of the PGI-D categories is different so max-score normalization is used to achieve comparability among indicators:

$$T_{dc} = \frac{\text{score}_{dc}}{\max_c}$$

where score_{dc} denotes the score of district d in category c , and $\max_c$ represents the maximum attainable score for that category. This normalization converts all category scores into a relative achievement scale between 0 and 1.

Falsity (F): Peer-Relative Deficit.

Falsity is the level of under-achievement against the national peer average of a given category. Rather than considering falsity as the converse of truth (as in fuzzy systems), falsity is represented as a peer-relative discrepancy in performance:

$$F_{dc} = \max(0, \bar{T}_c - T_{dc})$$

where $\bar{T}_c$ denotes the average normalized score across all districts for category c .

Indeterminacy (I): Structural Imbalance

Intra-district instability: models of indeterminacy cut across the educational dimensions. Development of some districts can be balanced with respect to indicators whereas at the same time others may be strong in certain areas but weak in others.

To measure this imbalance, a measure of indeterminacy within a district, the variance of normalized category scores is defined:

$$I_d = \text{Var}(T_{d1}, \dots, T_{d6})$$

The indeterminacy measure that was chosen is variance since it measures the dispersion on educational dimensions. Districts in which there is uniformity in the performance in the categories are characterized by low variance, which suggests structural stability, but, on the other hand, districts where the performance is not uniform are characterized by high variance, which depicts structural imbalance.

3. Feature Expansion

The neutrosophic representation yields:

1. 6 Truth components
2. 6 Falsity components
3. 6 Indeterminacy components, after replicating the Indeterminacy for all 6 categories using I_d

c1 (T,I,F)	c2 (T,I,F)	c3 (T,I,F)	c4 (T,I,F)	c5 (T,I,F)	c6 (T,I,F)
(0.441, 0.174, 0.000)	(0.778, 0.174, 0.000)	(0.627, 0.174, 0.000)	(0.686, 0.174, 0.000)	(0.500, 0.174, 0.000)	(0.571, 0.174, 0.002)
(0.462, 0.371, 0.000)	(0.778, 0.371, 0.000)	(0.608, 0.371, 0.002)	(0.743, 0.371, 0.000)	(0.340, 0.371, 0.000)	(0.488, 0.371, 0.086)
(0.397, 0.581, 0.000)	(0.756, 0.581, 0.000)	(0.706, 0.581, 0.000)	(0.771, 0.581, 0.000)	(0.340, 0.581, 0.000)	(0.357, 0.581, 0.217)
(0.424, 0.352, 0.000)	(0.700, 0.352, 0.000)	(0.843, 0.352, 0.000)	(0.714, 0.352, 0.000)	(0.480, 0.352, 0.000)	(0.774, 0.352, 0.000)
(0.414, 0.303, 0.000)	(0.711, 0.303, 0.000)	(0.863, 0.303, 0.000)	(0.714, 0.303, 0.000)	(0.560, 0.303, 0.000)	(0.726, 0.303, 0.000)

Fig. 3. T,I,F features of categories (first 5 rows)

leading to 18 features per district after we replicate I_d for consistency (13-D Independent features).

This transformation provides the expansion of the initial 6 dimensional performance space into uncertainty aware representation, which simultaneously measures the strength of achievement, structural deficit and the imbalance within the district.

4. Clustering Framework

In order to examine the impact of neutrosophic transformation on clustering behavior two similarity geometries were considered:

1. Euclidean K-means clustering, which clusters the districts based on the minimization of the within-cluster variance [3].
2. The cosine-based agglomerative clustering, which clusters districts by the similarity between feature vectors in terms of direction.

Separate clustering was done on:

1. the 6-dimensional normalized representation by the classical method
2. the neutrosophic representation (18 features).

Clusters validity was assessed by means of:

1. Silhouette Score, a cluster separability measure.
2. Davies-Bouldin Index, which is used to determine the compactness of the clusters.

4 Results and Conclusion

Performance clustering was measured using the Euclidean K-means and the cosine based similarity in the case of classical and neutrosophic representation.

Euclidean Clustering: The classical representation under Euclidean K-means scored a Silhouette Score of 0.314 and Davies-Bouldin (DB) Index of

1.155. The neutrosophic representation gave a Silhouette Score of 0.304 and a smaller DB Index of 1.115 which was slightly more compact cluster although with a slight change of silhouette.

Significantly, after neutrosophic transformation, 351 districts (45.7%) shifted cluster assignment. The difference in the average indeterminacy (0.410) in districts that changed clusters was much higher than in districts that did not change (0.268). This would indicate that Euclidean neutrosophic clustering primarily re-organizes a district that has a higher intra-district imbalance.

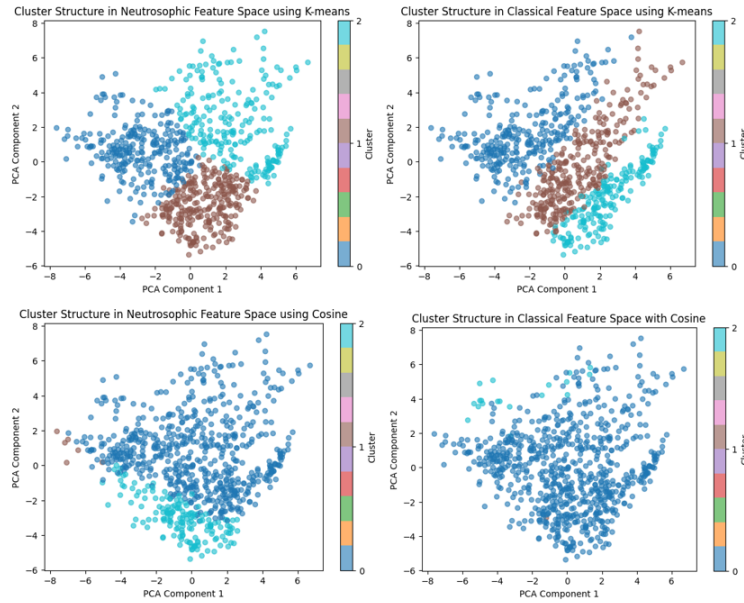


Fig. 4. clustering geometry-specific comparison Classical vs Neutrosophic

Cosine-Based Clustering: Under cosine similarity, classical clustering achieved a Silhouette Score of 0.269. Silhouette Score increased significantly to 0.389 with the neutrosophic representation which is an increase in separability at pattern level.

Under cosine-based neutrosophic clustering a total of 188 districts (24.5%) changed clusters. Unlike in the Euclidean case, the average indeterminacy in changing districts (0.161) was lower than in the stable ones (0.389). This indicates that cosine similarity interacts differently with the neutrosophic feature space, highlighting directional similarity, but not similarity of magnitude.

The findings reveal that neutrosophic expansion of features is geometrical. In the case of Euclidean distance, high-indeterminacy districts have higher chances

<i>K-means</i>			
Representation	Silhouette	DB Index	Reassigned (%)
Classical	0.314	1.155	—
Neutrosophic	0.304	1.115	45.7
<i>Cosine</i>			
Representation	Silhouette	Reassigned (%)	
Classical	0.269	—	
Neutrosophic	0.389	24.5	

Fig. 5. Euclidean and Cosine comparison Classical vs Neutrosophic

of being reorganized. Separability is enhanced significantly under cosine similarity and this implies that directional similarity is more effective with the T, I, F representation to discriminate against structure.

And through Figure 4 we can visually observe that clustering when done with Neutrosophication gives clearer clusters in both cases (K-means and Cosine-Based). In fact, classical clustering sorts almost all the clusters in 1 cluster in case of Cosine-Based. These visual results are a clear indication of significance of including the factor of Indeterminacy when needed, as in this dataset. A concise and interpretable summary of clusters (using Neutrosophic Logic) is given below.

Table 2. Educational interpretation of district clusters under neutrosophic clustering

Method	Cl.	District Count	Educational Profile	Interpretation
Cosine	0	599	Moderate scores across categories	Average-performing districts with balanced educational indicators
Cosine	1	6	Very low scores across indicators	Severely underperforming districts with systemic deficiencies
Cosine	2	16	Highest scores in most categories	High-performing districts with strong outcomes and digital readiness
K-means	0	269	Low digital learning and governance	Infrastructure developing but digital adoption remains weak
K-means	1	274	Highest scores across indicators	Strong performing districts with balanced educational development
K-means	2	225	Moderate outcomes with stronger infrastructure	Transitional districts with uneven educational development

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